

SMART BLIND CANE

Updated Components & Features List — ESP32 Platform | 2025

Note: HC-SR04 replaces JSN-SR04T ultrasonic sensors. Vibration motor (haptic feedback) and bone conduction headset references have been removed.

Hardware Components

#	Component	Purpose	Est. Cost (INR)
1	ESP32 DevKit v1	Central microcontroller + Bluetooth + WiFi	₹350
2	HC-SR04 Ultrasonic x2	Forward obstacle detection (dual mount)	₹160
3	IR Sensor Module	Pit and step-down detection	₹80
4	NEO-6M GPS Module	Real-time location acquisition	₹450
5	SIM800L GSM Module	SOS SMS to guardian	₹550
6	MPU6050 Accelerometer	Fall detection	₹150
7	DFPlayer Mini + MicroSD	Audio file playback	₹200
8	Buzzer	Audio alert output	₹30
9	Water / Rain Sensor	Puddle and wet surface detection	₹50
10	SOS Push Button	Manual emergency trigger	₹20
11	LiPo 3.7V 5000mAh	Primary rechargeable battery	₹750
12	TP4056 USB-C Charger	Battery charging management	₹40
13	MT3608 Boost Converter	Stable 5V supply to all modules	₹40
14	PVC Pipe + Grip	Cane body and handle	₹200
15	PVC Junction Box	Handle electronics enclosure	₹50
16	Misc (PCB, wires, etc.)	Assembly and prototyping	₹200
	TOTAL		~₹3,320

* HC-SR04 (non-waterproof) replaces JSN-SR04T. Recommended for indoor/sheltered use or with DIY waterproofing. Cost reduced from ₹440 to ₹160.

System Features

#	Feature	Description
1	Ultrasonic Obstacle Detection	Dual HC-SR04 sensors — one forward (chest height), one angled 30° downward. 5-sample
2	IR Pit / Step Detection	IR sensor at cane tip detects absence of ground reflection — flags stairs, pit edges, and ker

#	Feature	Description
3	Water / Puddle Detection	Resistive water sensor at tip detects surface moisture via conductivity. Fires voice alert 'We
4	Fall Detection	MPU6050 accelerometer monitors cane orientation. Sudden high-G event (>2.5G) followed
5	GPS Location Tracking	NEO-6M GPS module acquires live coordinates within 5 seconds upon SOS trigger. Duty-c
6	GSM SOS SMS Alert	SIM800L sends formatted SMS to guardian: 'EMERGENCY ALERT: [Name] needs help. Lo
7	Bluetooth A2DP Audio Alerts	ESP32 streams MP3 voice alerts directly to any paired Bluetooth earpiece — no extra hard
8	Priority-Based Alert System	5-level priority: (1) Pit/step drop — highest, (2) Obstacle <50 cm, (3) Obstacle <100 cm, (4)
9	Buzzer Audio Feedback	Onboard buzzer provides secondary audio cue for hazard alerts as backup to Bluetooth ear
10	Firmware Power Optimisation	Three strategies: ESP32 light sleep between reads (240 mA → 0.8 mA), GPS duty cycling,

SDG Alignment: SDG 3 (Good Health) · SDG 10 (Reduced Inequalities) · SDG 11 (Sustainable Cities) | Total Budget: ~■3,320 | Platform: ESP32 | Duration: 8 Weeks